# Multilevel phylogenetic inference of syntactic harmony in Indo-European

Yingqi Jing* (Institute of Modern Languages and Linguistics, Fudan University)

yingqijing@fudan.edu.cn

Joakim Nivre (Department of Linguistics and Philology, Uppsala University)

joakim.nivre@lingfil.uu.se

Michael Dunn (Department of Linguistics and Philology, Uppsala University)

michael.dunn@lingfil.uu.se

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* Corresponding author: yingqijing@fudan.edu.cn

**Abstract**

One of the most well-established typological universals is that languages tend to order the grammatical head and its dependents consistently. This harmonic principle has been attributed to cognitive benefits that facilitate language processing and learning efficiency. However, there is little corpus-based research disentangling the evolutionary processes that shape the crosslinguistic distribution of harmony. Using 58 dependency-annotated corpora and Bayesian multilevel phylogenetic inference, this study investigates the evolutionary biases towards harmony against principled baselines in the history of Indo-European. While the observed evolutionary trends favor harmony over disharmony, they are largely consistent with those expected under a random baseline that generates orders independently for each dependency type. At the same time, the observed trends show a much weaker harmonic bias than expected under a baseline that assumes a single head-dependent parameter governing word order patterns. Overall, these findings provide limited support for functional explanations of syntactic harmony in language use, while suggesting that distinct processes may play a role in shaping structural diversity and language evolution.

**Keywords:** multilevel phylogenetic inference; syntactic harmony; Universal Dependencies

# 1 Introduction

Linguistic typology has long observed that languages tend to order the grammatical head and its dependents consistently (Dryer 1992; Greenberg 1963; Hawkins 1983). For example, verb-object languages tend to be prepositional, while object-verb languages tend to be postpositional. Over the past decades, competing theories and new empirical findings have been advanced to explain the word order universals (also called "harmony"). These theories range from functional explanations focusing on cognitive and learning biases (Culbertson et al. 2012; Futrell et al. 2020; Hawkins 1983), to alternative views, emphasizing the roles of cultural evolution and grammaticalization in language change (Bybee 1988; Dunn et al. 2011).

On the one hand, syntactic harmony can be attributed to the principle of dependency locality or simplicity bias in learning. Research on word order distributions has shown that consistent head direction is crosslinguistically preferred, as it minimizes dependency distances between syntactically linked elements (Ferrer i Cancho 2004; Futrell et al. 2015; Gibson 1998; Gildea and Temperley 2010; Liu 2008). For example, if two phrases are hierarchically nested, consistent ordering (e.g., *cross* [*the border* [*of the country*]]) yields shorter dependency distances than inconsistent ordering (e.g., *cross* [[*the country's*] *border*]), where the intervening phrase *the country's* increases the distance between the verb *cross* and the object *border*. In parallel structures, some short dependents such as adjectives may depart from harmony to reduce dependency distance (Dryer 1992; Hawkins 1994; Temperley 2008; Temperley and Gildea 2017). Experimental evidence shows that shorter dependencies reduce working memory load, thereby facilitating sentence comprehension and production. For detailed reviews of dependency locality and its relationship to word order typology, see Liu et al. (2017), Temperley and Gildea (2017), Futrell et al. (2020), and Jing et al. (2022). Additionally, consistency in head direction can be motivated by simplicity bias in learning. Artificial language learning experiments have revealed that syntactic harmony enhances learning efficiency; in particular, learners prefer harmonic over disharmonic ordering of numerals and adjectives within noun phrases (Culbertson and Newport 2015; Culbertson et al. 2012, 2020). A grammar with

consistent head direction only requires learning a single parameter, while category-specific parameters are acquired for inconsistent structures (Baker 2001; Hsu and Chater 2010). See also Wang et al. (2025) for some recent experimental research on word order harmony between verb phrases and adpositional phrases, as well as Mansfield and Krapp (2025) for an alternative explanation of harmony in parallel dependents based on placing the head at the edge of the phrase.

On the other hand, syntactic harmony can be the product of cultural evolution and grammaticalization during language diversification and change. Dunn et al. (2011) adopted Bayesian phylogenetic models to detect correlated evolution across major word orders within four language families. Although some correlations are identified, they vary considerably across lineages and may largely reflect lineage-specific patterns rather than universal cognitive constraints (Levinson et al. 2011). See Jäger and Wahle (2021) and Hartung et al. (2022) for different findings and opinions about lineage-specific trends in word order universals, as well as a recent large-scale evolutionary analysis of 191 linguistic universals worldwide by Verkerk et al. (2025). Moreover, word order universals can be independently explained by grammaticalization, where certain syntactic categories share a common origin (Bybee 1988; Collins 2019; Cristofaro 2019). For instance, adpositions can derive from serial verb constructions through grammaticalization (Aristar 1991; Givón 1975). This naturally leads to consistent ordering between adpositional and verb phrases, without invoking functional explanations.

Leaving aside these competing theories and findings, most comparative studies on syntactic harmony rely on linguistic databases, such as the World Atlas of Language Structures (WALS; Dryer and Haspelmath 2013). Recent corpus studies are largely synchronic and show a distributional preference for harmony, which is often attributed to pressures for communication efficiency (Gerdes et al. 2019; Hahn et al. 2020; Naranjo and Becker 2018). However, it remains unclear whether synchronic distributions can adequately represent the evolutionary bias of syntactic change (Cysouw 2011; Maslova 2000; Widmer et al. 2017). For instance, although SOV order is globally dominant, large-scale comparative analyses show that changes from SOV to SVO are crosslinguistically more preferred than the reverse, favoring a disharmonic combination of SV and VO orders (Gell-Mann and Ruhlen 2011; Jing et al. 2023; Maurits and Griffiths 2014). Crucially, even though two word orders change towards the same direction, they can still evolve independently. Without explicit evolutionary modeling, it is difficult to determine whether there is any correlated evolution between word orders, that is, whether a change in one order (e.g., VO $\rightarrow$ OV) triggers a corresponding change in another (e.g., VS $\rightarrow$ SV; Levinson et al. 2011).

To address these issues, we investigate the evolutionary dynamics of syntactic harmony in Indo-European, using 58 language corpora from Universal Dependencies (UD) v2.15 (Zeman et al. 2024), and recent Indo-European language phylogenies (Heggarty et al. 2023). Following previous comparative work (Dunn et al. 2011; Hartung et al. 2022; Jäger and Wahle 2021; Levinson et al. 2011; Pagel 1994; Verkerk et al. 2025), we model the evolution of syntactic harmony as discrete states (harmony and disharmony) in the history of language phylogenies, assuming a continuous-time Markov chain (CTMC) model. To control for potential confounds such as word order regularization and independent change, we introduce two random baselines: one preserving the overall head direction of a language, and another maintaining the order of each dependency type in a language (see Section 2.2 for a detailed description of these baselines). The primary goal of this study is to detect additional evolutionary biases towards harmony beyond what is expected from these baselines. To this end, we integrate corpus-based analyses with multilevel phylogenetic inference to investigate the evolutionary mechanisms shaping word order patterns.

One key advantage of this approach is that it allows us to test hypotheses about word order

harmony in language use, as language change is ultimately driven by usage patterns. While previous comparative studies often treat word order probabilities as discrete values of 0 or 1, this coding strategy cannot accurately represent word order variation in actual utterances. For example, the order of adjective and noun in French is coded as postnominal in WALS (Dryer and Haspelmath 2013), whereas it is described as both prenominal and postnominal in Grambank (Skirgård et al. 2023). According to the UD French corpus, there are approximately 70% NAdj utterances (or 30% AdjN instances). This suggests that standard typological databases may underestimate word order variation in certain contexts while overestimating it in others (Jing et al. 2023). Here we adopt a corpus-based approach by estimating the joint probability distribution of word orders within sentences, allowing these probabilities to range over the entire interval between 0 and 1(see details in Section 2.2).

Furthermore, phylogenetic modeling offers a promising method for detecting evolutionary biases towards or against harmony in the history of language families (Dunn et al. 2011; Levinson et al. 2011). We take an initial step by focusing on Indo-European languages, but extend beyond earlier phylogenetic methods that analyze character changes separately for individual word orders or word order pairs (Dunn et al. 2011; Hartung et al. 2022; Jäger and Wahle 2021; Jing et al. 2023; Maurits and Griffiths 2014). Here we develop a multilevel phylogenetic model that simultaneously estimates the overall rates between states and specific rates for individual word order pairs. This model is more robust in parameter estimates than separate analyses of correlated character changes, as it allows rate estimates to be jointly informed and partially pooled across word order pairs (McElreath 2020; Nalborczyk et al. 2019; Stan Development Team 2023). Further details of the multilevel phylogenetic modeling are given in Section 2.3 and supplementary materials S3.

This study assesses the impact of functional pressures on word order change, focusing specifically on pairwise word orders observed in actual utterances, in line with prior research in linguistic typology, comparative linguistics, and language processing (Dunn et al. 2011; Gibson 1998; Hawkins 1983; Jäger and Wahle 2021). We leave it to future research to investigate lineage-specific patterns and grammaticalization pathways in language change.

# 2 Data and methods

## 2.1 Universal Dependencies and Indo-European language phylogenies

We use dependency-annotated corpora from UD v2.15 (Zeman et al. 2024). All UD corpora have been standardized into a consistent annotation scheme (de Marneffe and Manning 2008; de Marneffe and Nivre 2019; de Marneffe et al. 2021; Petrov et al. 2012). To avoid ongoing controversies about the status of function words (Osborne and Gerdes 2019; Osborne and Maxwell 2015), we focus exclusively on dependencies between major lexical categories, such as verb, noun, adjective, and adverb. This gives us 12 major dependency types, as shown in Table 1. The choice of these dependencies guarantees a sufficient number of tokens in each corpus (on average $n \geq 1000$). See detailed summaries in Table S1 in the supplementary materials.

Table 1: Summary of 12 major dependency types from the UD database.

| Dependency type | Notation | Example |
|---|---|---|
| verb-nominal subject | V-S | *the girl ran away* |
| verb-object | V-O | *read the book* |
| verb-oblique | V-Obl | *go to the park* |
| noun-adjectival modifier | N-Adj | *a nice guy* |
| noun-nominal modifier | N-Nmod | *the cover of the book* |
| verb-adverbial modifier | V-Adv | *walk slowly* |
| noun-adverbial modifier | N-Adv | *only one choice* |
| adjective-adverbial modifier | Adj-Adv | *very tasty* |
| verb-open clausal complement | V-Xcomp | *he wanted to eat an apple* |
| verb-clausal complement | V-Ccomp | *he said that he knew it* |
| verb-adverbial clause | V-Advcl | *she was reading when I came back* |
| noun-adnominal clause | N-Acl | *the dish you like* |

We use the latest phylogenetic trees of Indo-European from Heggarty et al. (2023). The UD corpora include several ancient languages, such as Middle French, Old French, and Old East Slavic, which are not covered in the phylogenies. We graft them onto the phylogenies according to conventional assumptions about the relevant branching events (Jing et al. 2023; Widmer et al. 2017). Additionally, we incorporate two contemporary languages, Serbian and Pomak, by attaching them to their closest relatives (Croatian and Bulgarian, respectively) in the phylogenies. This provides us with 58 Indo-European languages mapped onto the phylogeny.

## 2.2 Measuring syntactic harmony

We quantify syntactic harmony by assessing whether pairs of dependency types have the same order within sentences – both head-initial or both head-final. This parallels typological analyses of word order correlations, but focuses on co-occurrence patterns in actual utterances rather than distributions across languages. These two perspectives coincide when a language has a strictly fixed word order. However, since traditional categorical coding of word orders cannot fully capture the variation in language use, we adopt a probabilistic measure of harmony based on the co-occurrence of word orders within sentences. This allows a more precise assessment of the cognitive benefits of harmony in language use, as language processing involves identifying dependencies and integrating linguistic units to build the structure of a sentence. If harmony is functionally motivated, such effects should be detectable in how word orders co-occur within sentences.

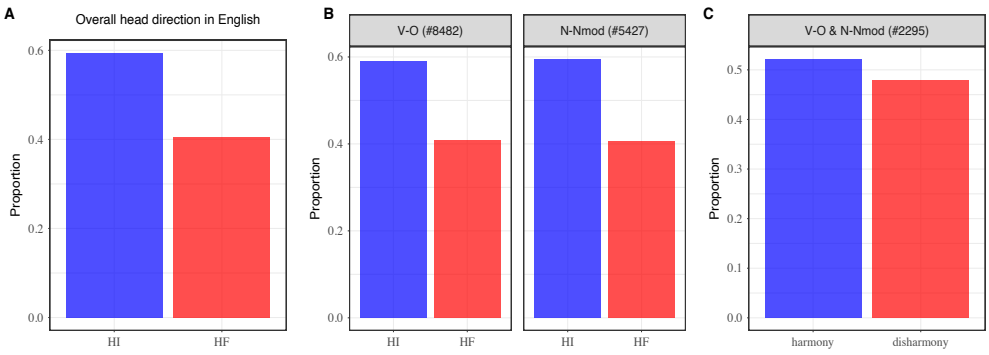
For each language, we compute observed frequencies of harmonic and disharmonic combinations of word orders within sentences. To account for variation in frequency and corpus size, we enter the raw frequencies into a Bayesian binomial model to estimate the probabilities of (dis)harmony and sample from the posterior to incorporate uncertainty in our estimates (Jing et al. 2023). However, these probabilities should not be directly interpreted as evidence for or against harmony, since it is unclear whether they faithfully reflect underlying cognitive or evolutionary biases (Cysouw 2011; Maslova 2000; Widmer et al. 2017), or instead arise from other processes, such as word order regularization or independent change. One factor is word order regularization (Culbertson et al. 2012; Hawkins 1983; Jing et al. 2021). The order of a language can become more regular

to facilitate learning or communication efficiency, which increases overall consistency without correlated evolution. Similarly, the distributional word order patterns can be driven by independent evolution. Suppose a language initially has 70% VO and 90% NAcl. Over time, the proportion of VO order increases to 80%, while NAcl order remains stable. Although this raises the harmony between VO and NAcl, there is no evidence for correlated evolution (Ladd et al. 2015; Levinson et al. 2011). Previous typological studies have primarily focused on distributional preferences of word orders without distinguishing alternative forces. To accurately evaluate the harmonic bias while minimizing confounding effects, we control for the base distributions of word orders in each language by introducing two baselines that randomly generate orders for individual dependency types in a language.

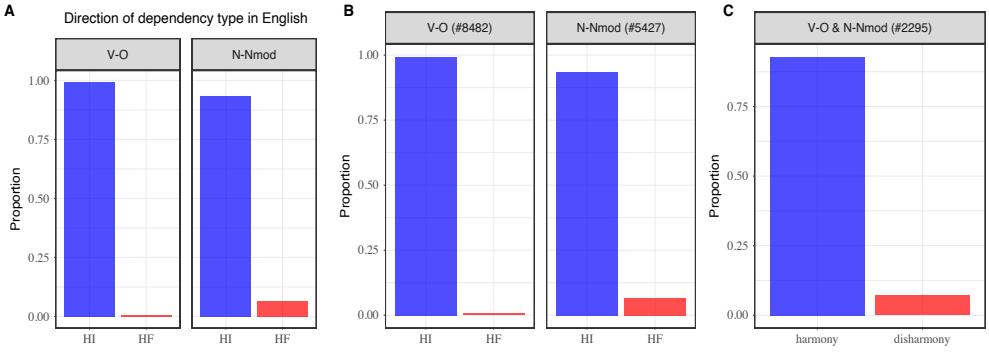
The first random baseline is designed to preserve the overall head direction in a language, regardless of dependency type. Under this baseline, we randomly assign an order for each dependency type in a sentence according to the overall head direction estimated from the text corpora. This is illustrated in Figure 1a, where English has approximately 60% head-initial (HI) and 40% head-final (HF) dependencies. We randomly assign an order for each dependency type based on this proportion. The harmony of two dependencies – in the case of Figure 1a, V-O and N-Nmod – is calculated by counting observations when they co-occur in a sentence with the same order. In this case, the two dependency types have a 52% probability of harmony and a 48% probability of disharmony. This baseline assumes a single probabilistic distribution governing the order of all dependency types. Speakers or learners would only need to acquire one head-dependent direction to produce the orders of different dependencies. Theoretically, this baseline approximates the hypothesis of head-dependent parameters that ultimately underlie the simplicity bias in language learning.

The second random baseline is generated by randomly assigning an order to each dependency type while preserving the proportions of individual dependency types in a language. For each dependency type in a language, we first summarize its word order distributions. For instance, in English, 99% of verbs precede objects (VO), and 93% of nouns precede the nominal modifiers (NNmod), as shown in Figure 1b. These proportions serve as base probabilities for generating random orders for each dependency type. We then count the co-occurrence of word orders within sentences to compute the proportions of harmony and disharmony. Because the order of each dependency type is drawn independently, their joint probability is approximately the product of the individual probabilities, with some random variation arising from differences in frequency and corpus size. This baseline can be interpreted as a model in which speakers learn and produce each dependency type independently; that is, the order of one dependency type does not influence the order of another (Jing et al. 2022).

Our primary goal is to assess additional evolutionary bias towards harmony beyond what is expected from the random baselines. If two orders are constrained by harmony, they are expected to evolve in a correlated manner: a change in one order will trigger a corresponding change in another (Dunn et al. 2011; Levinson et al. 2011; Pagel 1994). Both baselines are generated independently for each dependency type without correlated change between dependency types. The key distinction lies in the base distributions used to generate random orders. In the first baseline, word orders are drawn from a single probabilistic distribution across all dependency types. In the second baseline, by contrast, orders are drawn from category-specific distributions, allowing each dependency type to vary independently. Crucially, the randomized data preserve the exact number of co-occurrences for each word order pair in the text corpora, ensuring a fair comparison and minimizing potential biases. We generate 200 random permutations per dependency type per language, and evaluate the observed harmony against these baselines.




(a) Random baseline 1




(b) Random baseline 2

Figure 1: Graphic representations of the procedures for generating random baselines: (a) random baseline 1; (b) random baseline 2. The numbers following the dependency types indicate the observed frequencies of individual word orders or word order pairs from the English corpus.

## 2.3 Multilevel phylogenetic modeling

We use Bayesian phylogenetic inference to test hypotheses about word order harmony in language use along the history of Indo-European, while controlling for the base distributions of individual word orders in each language. Unlike previous corpus-based studies that focus on the distributional preferences of word orders (Gerdes et al. 2021; Hahn et al. 2020; Naranjo and Becker 2018), phylogenetic approaches seek to understand the evolutionary processes that generate the joint probability distributions of word orders in Indo-European. We fit a two-state CTMC model on an explicit phylogeny to estimate the rates of change between states (i.e., $q_{12}$: rate of change from disharmony to harmony; $q_{21}$: rate of change from harmony to disharmony). We also go beyond previous phylogenetic approaches that analyze the correlated evolution of each word order pair separately (Dunn et al. 2011; Hartung et al. 2022; Jäger and Wahle 2021). Instead, we develop a multilevel phylogenetic CTMC model that can not only capture the overall transition rates between states, but also rate variation across individual word order pairs (66 pairs in total), guarding against overfitting and underfitting (Nalborczyk et al. 2019; Stan Development Team 2023). Full model specification and implementation details are provided in the supplementary materials S3.

We fit the multilevel CTMC model separately for the observed data and two random baselines

via the **rstan** package in R (Hoffman and Gelman 2014; Stan Development Team 2023). The strength of evolutionary biases towards harmony is assessed by computing the posterior rate ratio between the forward and backward rates ($q_{12}/q_{21}$), and comparing it to the corresponding random baselines. If functional pressures do shape grammars towards consistency in head direction in actual utterances, we would expect to observe a higher rate ratio for harmony when compared to the random baselines.

# 3 Results

Figure 2 shows the posterior distributions of forward and backward rates ($q_{12}$: rate of change for harmony; $q_{21}$: rate of change for disharmony) and their rate ratios ($q_{12}/q_{21}$) for the observed data and two random baselines at the population level. Regardless of the direction of changes, our results show broad overlaps between the observed distribution and the second random baseline, suggesting no clear rate differences (see Tables S3–S4 for detailed summaries in the supplementary materials). Their rate ratios are also virtually identical (mean $\text{ratio}_{\text{obs}} = 1.99$, 95% CI = [1.01, 2.91]; mean $\text{ratio}_{\text{rand2}} = 1.87$, 95% CI = [0.9, 2.78]; $p(\text{ratio}_{\text{obs}} < \text{ratio}_{\text{rand2}}) = 0.43$). By contrast, there is a marked difference in rate estimates when compared to the first baseline. The observed rate for harmony is notably lower (mean $\text{rate}_{\text{obs}}$: 1.06, 95% CI = [0.53, 1.55]; mean $\text{rate}_{\text{rand1}}$: 4.49, 95% CI = [2.96, 6.32], $p(\text{rate}_{\text{obs}} < \text{rate}_{\text{rand1}}) = 1$), while the rate for disharmony is relatively higher (mean $\text{rate}_{\text{obs}}$: -0.93, 95% CI = [-1.37, -0.48]; mean $\text{rate}_{\text{rand1}}$: -1.84, 95% CI = [-3.4, -0.38]; $p(\text{rate}_{\text{obs}} > \text{rate}_{\text{rand1}}) = 0.84$). This substantially reduces the observed rate ratio relative to the first baseline, indicating a much weaker evolutionary bias towards harmony than expected from the first baseline (mean $\text{ratio}_{\text{rand1}} = 6.32$, 95% CI = [4.87, 8.21]; $p(\text{ratio}_{\text{obs}} < \text{ratio}_{\text{rand1}}) = 1$). It is also worth noting that the log rate ratios are well above 0 ($p(\text{ratio} > 0) = 1$) for both the observed results and random baselines in Figure 2 and Table 2, indicating a strong evolutionary tendency towards harmony over disharmony. This also aligns with our expectation that the general consistency in word order can arise from alternative processes.

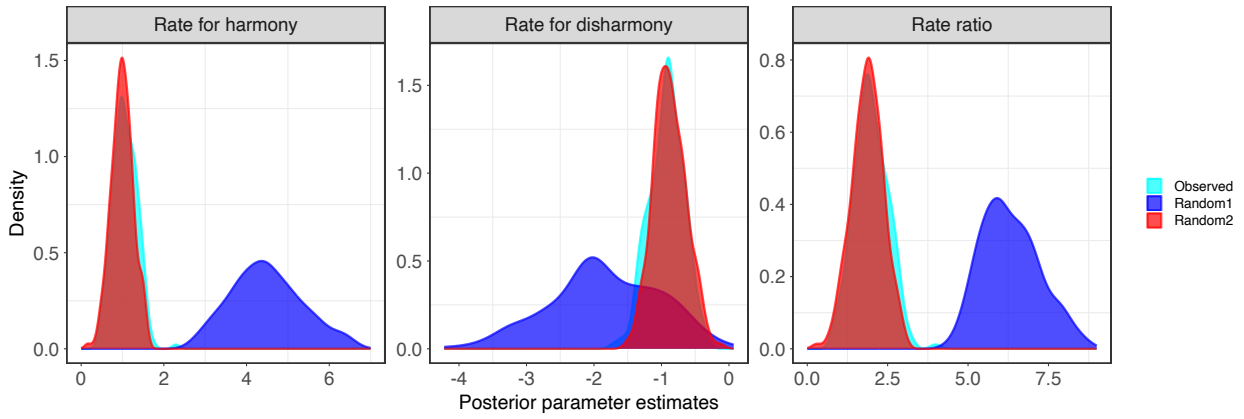



Figure 2: Distribution of the forward rates for harmony ($q_{12}$), backward rates for disharmony ($q_{21}$), and their rate ratios ($q_{12}/q_{21}$) for the observed data and two random baselines at the population level ("fixed effects"). The estimated rates and rate ratios are displayed on a log scale. A higher log rate ratio indicates a stronger evolutionary biases towards harmony, while a lower log rate ratio suggests an evolutionary bias towards disharmony. The dashed line in the right panel indicates a log rate ratio of 0, representing equal rates.

Table 2: Summary of estimated rates of change and rate ratios between observed data and two random baselines at the population level. The estimated rates and rate ratios are displayed on a log scale.

| Type | Parameter | Mean | 95% CI | $p(\text{value} > 0)$ |
|---|---|---|---|---|
| Observed | Rate for harmony | 1.06 | [0.53, 1.55] | — |
| | Rate for disharmony | -0.93 | [-1.37, -0.48] | — |
| | Rate ratio | 1.99 | [1.01, 2.91] | 1 |
| Random1 | Rate for harmony | 4.49 | [2.96, 6.32] | — |
| | Rate for disharmony | -1.84 | [-3.4, -0.38] | — |
| | Rate ratio | 6.32 | [4.87, 8.21] | 1 |
| Random2 | Rate for harmony | 0.99 | [0.49, 1.49] | — |
| | Rate for disharmony | -0.88 | [-1.31, -0.42] | — |
| | Rate ratio | 1.87 | [0.9, 2.78] | 1 |

Our model also allows us to assess the extent to which rates of change vary across individual word order pairs ("random effects"). Figure 3 shows the posterior estimates of log rate ratios for the observed and two random baselines among individual word order pairs. Overall, the random effects closely mirror the patterns observed at the population level. In particular, the observed distributions for all word order pairs show substantial overlaps with the second random baseline, with no evidence for additional evolutionary biases towards harmony. See Tables S11 and S12 in the supplementary materials for summary statistics of log rate ratios between the observed and the second random baseline.

When compared to the first random baseline, the observed rate ratios are clearly lower for most word order pairs, though there is appreciable variation among them. The most pronounced differences occur in adjectival modifiers, complement clauses, and adverbial modifiers of adjectives, which show substantially weaker harmonic bias across word order pairs. Subjects, objects, and adverbial modifiers of nouns also show noticeable divergence from the first baseline across seven word order pairs. A slightly weaker divergence is observed for nominal modifiers and adnominal clauses, which exhibit marked differences in six out of eleven word order pairs. This is followed by three other dependency types, including V-Obl, V-Adv, and V-Xcomp, and each shows notable differences with five other orders. Finally, adverbial clauses align most closely with the first baseline. The detailed comparisons of rate ratios for individual word order pairs between the observed data and the first random baseline can be found in Table S9 in the supplementary materials. Note that the estimated rate ratios for the first random baseline remain notably stable throughout (blue density distributions in Figure 3). This comes as no surprise, since all word order pairs are drawn from the same overall distribution of head direction within each language. While the observed frequencies vary across word order pairs and languages, the estimated rate ratios show little variation in the entire language family.

To further compare rate ratios between syntactic structures, we categorize word order pairs into three groups: verb-headed, noun-headed, and cross-category pairs. Verb-headed and noun-headed dependencies tend to be parallel, while cross-category pairs are more likely to be nested. Figure 4 visualizes only those pairs with a marked difference (red cells with an asterisk) from the first random baseline. Both parallel and cross-category pairs show substantial discrepancies, suggesting a systematic weaker harmonic bias relative to the first baseline. Specifically, subjects, complement

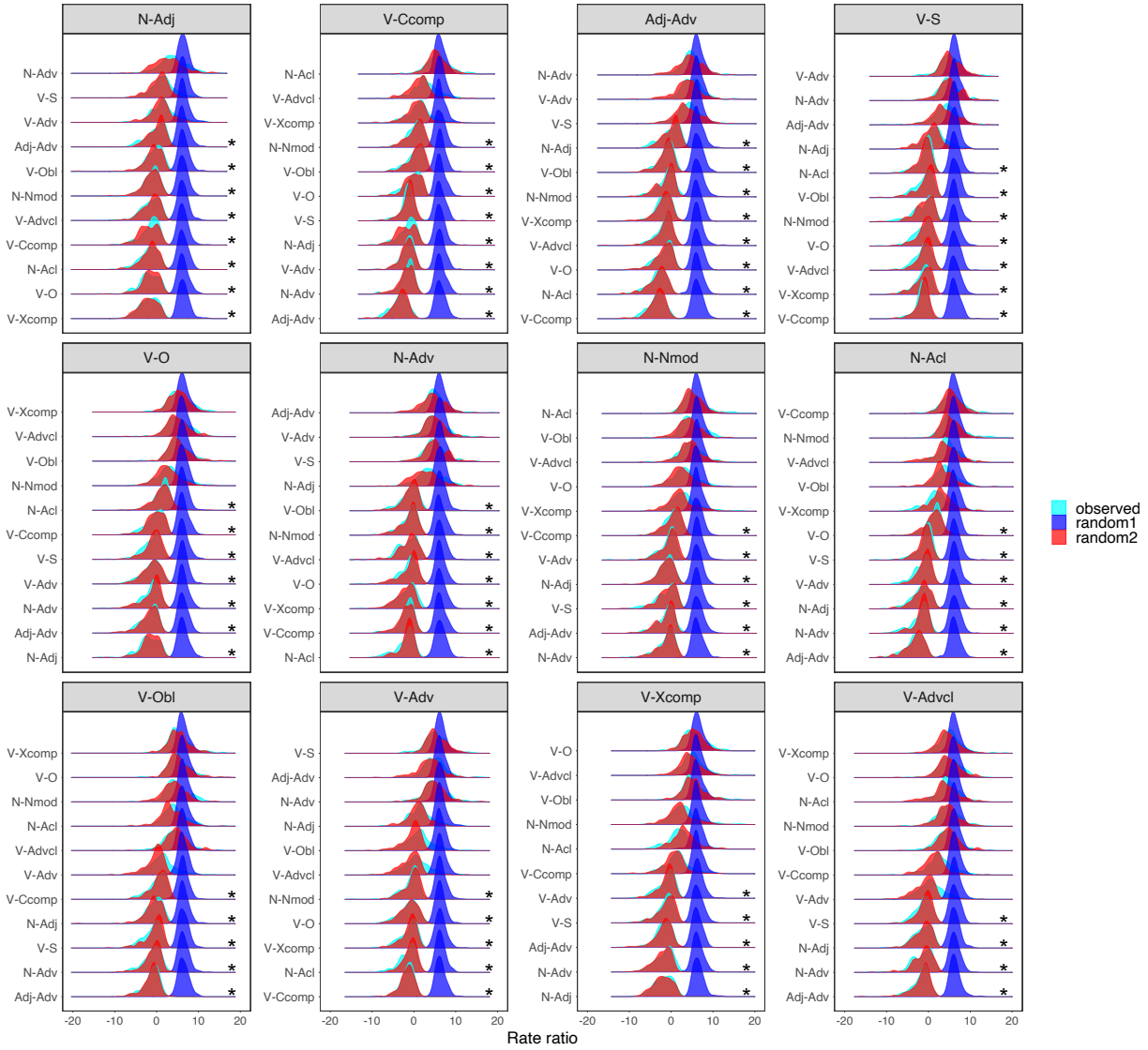



Figure 3: Distribution of log rate ratios between forward and backward rates for the observed and two random baselines across individual word order pairs ("random effects") with the Indo-European phylogenies from Heggarty et al. (2023). For easier interpretation, we visualize the posterior rate ratios by comparing each order to all other orders. In each panel, distributions are arranged based on the difference in rate ratios between the observed and the first random baseline. Word order pairs that closely match the first baseline are placed higher, while those with greater discrepancies are placed lower. We mark those pairs with a significant difference in rate ratio ($p < 0.05$) with an asterisk (*) on the right. These panels are further sorted by the number of marked word order pairs. Word orders that exhibit greater harmony with others (relative to the first random baseline) are placed in the top row, while those that are more disharmonic are placed in the bottom row.

clauses, and adverbial modifiers of verbs tend to be less harmonic with other verbal arguments or modifiers. Among nominal modifiers, adjectival modifiers of nouns and adverbial modifiers of nouns also show less harmony with other elements. Cross-category pairs deviate considerably from the first baseline as well; in particular, adverbial modifiers of adjectives are disharmonic with both verb-headed and noun-headed dependencies. Similar divergent patterns emerge when we restrict cross-category pairs to hierarchically nested categories in Figure S2 in the supplementary materials.

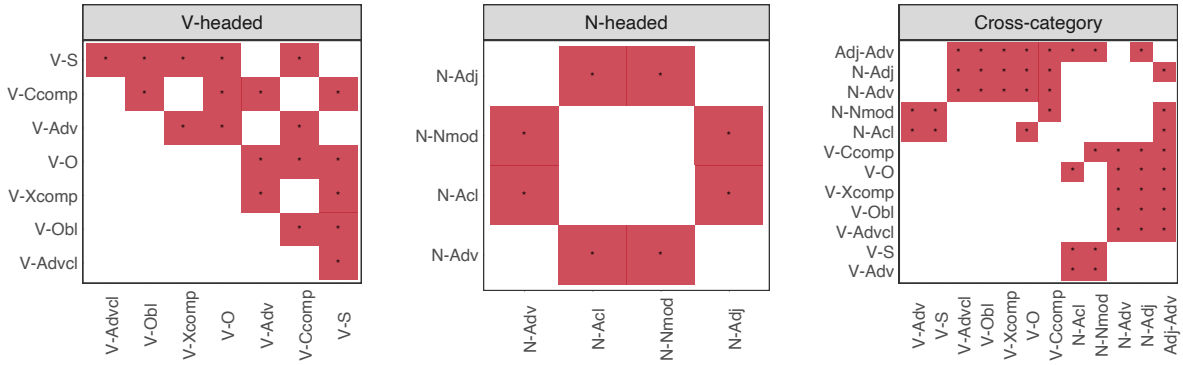



Figure 4: Heatmap of differences in rate ratio for verb-headed dependencies, noun-headed dependencies, and cross-category pairs, relative to the first random baseline. Red cells with an asterisk (*) indicate a significant difference in rate ratio ($p < 0.05$) compared to the first baseline.

# 4 Discussion and conclusions

Our findings do not align well with the functional explanations of syntactic harmony in comprehension, production, or learning (Culbertson et al. 2012; Futrell et al. 2020; Hawkins 1983). Although we observe an overall preference for harmony over disharmony (log rate ratios above 0 in Figure 2 and Table 2), there is no evidence for additional evolutionary biases towards harmony beyond what is expected under random baselines. Instead, the observed rates of change and rate ratios overlap broadly with the second random baseline. When word order distributions are held constant within languages, there is little room left for additional harmonic constraints across dependency types. In other words, the observed tendencies for co-occurrence of word orders can arise independently, without invoking cross-categorical harmony. This raises the question of why syntactic harmony has been previously reported (Gerdes et al. 2019; Hahn et al. 2020; Naranjo and Becker 2018). One possibility is that these distributional preferences cannot directly reveal the underlying processes that generate the current word order patterns. In particular, they cannot distinguish between correlated evolution, independent change, and general regularization, all of which can produce harmony. By explicitly comparing observed biases to principled baselines, we show that the usage patterns of word orders are broadly consistent with independent evolutionary processes specific to individual word orders, with limited evidence for additional cross-categorical harmony.

More importantly, our analysis reveals a much weaker harmonic bias compared to the first random baseline at the population level. This is primarily driven by a reduced rate for harmony and an increased rate for disharmony (Figure 2). This contrasts with prior work emphasizing the cognitive benefits of word order consistency. Our first random baseline reflects the idea of constructing

coherent grammars through a generalized head-dependent parameter. However, word orders in Indo-European seem to evolve towards more mixed configurations. This challenges the formal and functional theories that posit a single head-dependent parameter generalizable across dependency types to reduce grammatical complexity and facilitate language learning (Baker 2001; Culbertson et al. 2012; Hsu and Chater 2010). Rather, word order changes might be more diverse than commonly assumed (Evans and Levinson 2009), and different dependency types tend to regularize in distinct ways.

Our results also diverge from prior phylogenetic studies suggesting correlated word order evolution, albeit to varying degrees, across language families (Dunn et al. 2011; Hartung et al. 2022; Jäger and Wahle 2021; Verkerk et al. 2025). A full comparison is limited, as these studies adopt different measures and analyze word order pairs separately. We suspect that one key reason for this discrepancy lies in our approach: we measure word order harmony in actual utterances and assess the observed harmonic bias against principled baselines. This stands in contrast to previous studies that examine word order universals using categorical grammar data, which cannot fully capture the word order variation in language use (Jing et al. 2023; Levshina 2019). Crucially, the overall evolutionary patterns we observe only emerge in a multilevel phylogenetic framework, rather than through separate analyses of individual word order pairs. It remains unclear to what extent the word order correlations identified in earlier studies deviate from the overall evolutionary trends. Despite this, if we focus only on Indo-European languages, which show the most prominent results in Dunn et al. (2011), correlated changes are relatively rare (only 9 out of 28 possible pairs). If we move from grammar data to corpus analysis of word order patterns in actual utterances, one may find that word order changes might be more diverse than expected, with no sufficient evidence for correlated evolution in language use, as observed in our study.

Across individual word order pairs, the estimated log rate ratios vary considerably, but a consistent pattern holds: the observed ratios align closely with the second random baseline and diverge mostly from the first baseline. This suggests that the observed co-occurrence patterns between word orders are largely consistent with the patterns expected under independent processes, leaving limited evidence for additional cross-categorical harmony. By contrast, when compared to the first baseline, our results show a notably weaker harmonic bias across dependency types. This divergent trend cannot be fully explained by the dependency locality theory, which predicts greater disharmony in parallel than hierarchically nested structures (Gulordava 2018; Temperley 2008; Temperley and Gildea 2017). Under this view, verb-headed and noun-headed dependencies are expected to show more disharmony than cross-category pairs (in Figure 4). Yet our results show that parallel dependents (14 of 27, or 52%) display similar (or slightly lower) degrees of divergence compared to cross-category pairs (24 of 39, or 62% of pairs). We suspect that the locality effects may be more evident in specific word orders than in general patterns across syntactic categories. In particular, the divergence in verb-headed dependencies is likely linked to the directional change from SOV to SVO in Indo-European, where dependency locality may have driven the disharmonic combination of SV and VO (Futrell et al. 2015; Hawkins 1983; Jing et al. 2023; Liu 2008; Maurits and Griffiths 2014; Temperley and Gildea 2017). Moreover, verbal arguments are more susceptible to factors like topicalization and inversion, making them more variable. Similarly, adjectival modifiers and adverbial modifiers of adjectives depart substantially from the first baseline. These findings are in line with previous typological observations that short modifiers, such as adjectival modifiers of nouns and adverbial modifiers of adjectives, are more flexible and thus less likely to correlate with the order of V and O (Dryer 1992; Hawkins 1983), or they may be placed at the opposite side of the head to achieve dependency length minimization (Temperley 2008; Temperley and Gildea 2017).

Our results also suggest that complement clauses diverge substantially from harmony relative to

the first baseline. One possible source of divergence is that clausal complements in UD include both traditional complement clauses with a complementizer and reported speech, which introduces additional variation. This effect is especially pronounced given that many UD corpora are based on newspaper texts, where quoted speech is frequent and can occur either before or after the main verb. Another possibility is that in Indic languages (strictly head-final languages), complement clauses typically follow the verb, producing strong disharmony with other elements. Therefore, the divergence stems not only from structural diversity across dependency types, but also from effects of text genre and variation between language subfamilies. Together, these divergent patterns underscore the need for finer-grained corpus and experimental work on individual syntactic structures across language subfamilies.

# Supplementary materials

Code and data summaries are provided in the supplementary materials. The executable scripts, input data, and model results are available in an Open Science Foundation repository (https://osf.io/y4g3j).

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# Authors' contributions

YJ, JN, and MD contributed to the conceptualization and research design; YJ and MD developed the theoretical framework; YJ conducted the statistical analyses and drafted the manuscript; and all authors discussed the results and contributed to reviewing and editing the manuscript.

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